

One framework, five decisions, nine screens.



Everything needed to build Sidera in code: the stack and why it wins, the layer boundaries, the astrological domain contracts, the rule engine that writes each day's reading, the screen and route map, and the token set the interface is drawn from.

The design files stay the source of truth for appearance.
This document is the source of truth for structure.

Runtime	React + Vite PWA	Ephemeris	Swiss, WASM, on-device
Readings	Rule-based, deterministic	Storage	Local, sync-shaped
Network	None required	v1 scope	9 screens, no push

Choose the runtime the ephemeris already lives in

Swiss Ephemeris is C. Every platform reaches it through a binding, and the bindings differ enormously in cost. A browser runtime gets an official WASM build that runs in-process with no bridge, no native module, and no store review. That single fact decides the stack.

CANDIDATE	PATH TO SWISS EPHEMERIS	FRICTION
React + Vite	Official WASM build, loaded as a module. Ephemeris files served as static assets and cached by the service worker.	Lowest
React Native	No maintained JSI binding. Realistic options are a hidden WebView or writing your own native module twice.	High
Flutter	FFI to a compiled C library per platform. Workable, but you own the build toolchain for each target.	Medium
SwiftUI	Clean C interop and the best typography rendering, but one platform and a store review on every change.	Medium

DEPENDENCIES

vite · react · typescript	shell
swissep-wasm	ephemeris
zustand	state
dexie	IndexedDB
luxon	time zones
vite-plugin-pwa	offline
@capacitor/core	later, stores

CONSEQUENCES TO ACCEPT

Ephemeris files add roughly 2–5 MB to the install. Precache them; the app is unusable without them and they never change.

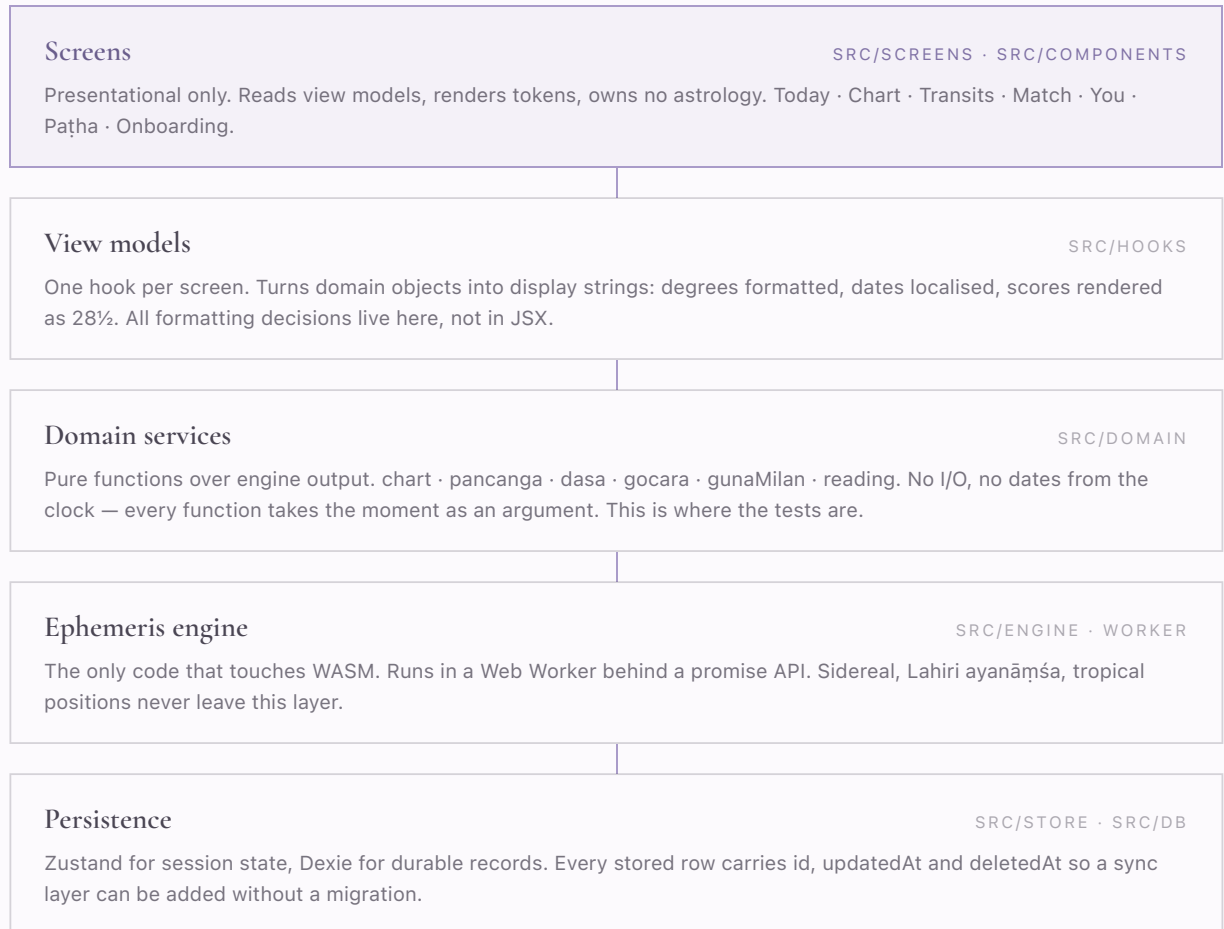
Run the WASM in a Web Worker. A full chart is fast, but a daśā tree or a year of transits will block the main thread if computed inline.

Local notifications need the Capacitor wrapper. That is the one v2 feature the PWA cannot do well, and it is out of v1 scope.

Keep every DOM-touching concern behind the component layer so the same core ships inside Capacitor unchanged.

Five layers, dependencies pointing one way

Each layer may import from the layer below it and never from the layer above. The engine knows nothing about React; the components know nothing about Swiss Ephemeris. This is the rule that keeps the astrology testable and the interface replaceable.



CACHING RULE

A natal chart is computed once per person and memoised for the session. Pañcāṅga and gocara are computed once per person per day, keyed personId:YYYY-MM-DD.

TESTING RULE

Snapshot the sample persona — Aisha Rao, 14 Aug 1998, 04:32, Jaipur — against known values: Siṃha lagna, Candra in Rohiṇī pada 2, Śani mahādaśā. Any drift is a regression.

File structure

src/		components/	
engine/		Kundli.tsx	north-indian diamond
sweph.worker.ts	WASM boot, raw calls	ChipRow.tsx	exclusive select
client.ts	promise API, transferable	Ledger.tsx	hairline row list
ayanamsa.ts	Lahiri offset	Segmented.tsx	rasi/navamsa/chalit
types.ts	engine-level shapes	UnderlineField.tsx	onboarding input
		GhostGlyph.tsx	decorative numeral
domain/		TabBar.tsx	five tabs
chart.ts	rasi, navamsa, chalit	ScreenHeader.tsx	name + meta + rule
houses.ts	bhava assignment		
pancanga.ts	tithi yoga karana paksa	store/	
dasa.ts	vimsottari tree	session.ts	active person, tab
gocara.ts	transit events, sadhe sati	settings.ts	palette, ayanamsa
gunaMilan.ts	8 kutas, 36 points	db.ts	Dexie schema v1
reading/			
detect.ts	chart -> conditions	theme/	
select.ts	weighted, seeded pick	palettes.ts	six token sets
compose.ts	fragments -> statement	type.ts	scale + families
fragments.ts	the authored library	space.ts	4px base
fixtures/			
aisha.ts	snapshot persona	public/	
		ephe/	.se1 files, precached
hooks/			
useToday.ts	statement, chip panels		
useNatalChart.ts	D-1 / D-9 / chalit		
useTransits.ts	week strip, ledger		
useMatch.ts	kuta rows, verdict		
useProfile.ts	settings, saved people		
screens/			
Today.tsx	Chart.tsx	Transits.tsx	
Match.tsx	You.tsx	Patha.tsx	
onboarding/Step1.tsx	Step2.tsx		

NAMING

Sanskrit terms keep their transliteration in the interface and lose their diacritics in code: pancanga.ts exports Pancanga, the screen prints *pañcāṅga*. One display dictionary in theme/terms.ts maps keys to their typeset forms so no diacritic is ever hand-typed in a component.

What the engine returns, and what the screens receive

Nine types carry the whole app. Fix these first — the screens, the reading engine and the tests all depend on their shape, and changing one later is expensive.

```

type BirthData = { name: string; utc: string; tzOffsetMin: number
                  lat: number; lon: number; place: string
                  timeKnown: boolean }

type Graha = { id: GrahaId; lon: number; rasi: RasiId; degInRasi: number
              nakshatra: NakshatraId; pada: 1|2|3|4
              bhava: number; retrograde: boolean; combust: boolean }

type Chart = { varga: 'D1'|'D9'|"chalit"; lagna: RasiId
              lagnaDeg: number; grahas: Record<GrahaId, Graha>
              bhavaLords: RasiId[] }           // 12, index 0 = 1st

type Pancanga = { tithi: { id: number; name: string; paksa: 'sukla'|"krsna"
                        endsAt: string }
                nakshatra: { id: NakshatraId; endsAt: string }
                yoga: string; karana: string
                sunrise: string; sunset: string; weekday: number }

type DasaNode = { lord: GrahaId; level: 1|2|3; from: string; to: string
                 children?: DasaNode[] }

type GocaraEvent = { at: string; graha: GrahaId
                    kind: 'ingress'|"retrograde"|"direct"|"combust"|"aspect"
                    fromRasi?: RasiId; toRasi?: RasiId
                    relativeToMoon: number }    // 1–12, for gocara strength

type SadheSati = { active: boolean; phase: 'rising'|"peak"|"setting"
                 yearOf: number; total: 7.5; startedAt: string }

type KutaScore = { kuta: KutaId; score: number; max: number; note?: string }

type GunaMilan = { a: BirthData; b: BirthData; rows: KutaScore[]
                  total: number; max: 36
                  mangalDosa: { a: boolean; b: boolean; cancelled: boolean }
                  verdict: string }

```

INVARIANTS

All longitudes sidereal and normalised to [0, 360). All instants stored as UTC ISO strings; local time is a rendering concern. Nakṣatra and pada derive from longitude only — never store them independently.

UNKNOWN BIRTH TIME

When timeKnown is false, assume noon local, set lagna to the Moon's rāśi (candra lagna), and suppress every bhāva-dependent claim in the reading and the daśā start date.

How the day gets its sentence

No network, no model, no randomness that isn't reproducible. The chart and the date go in; a fixed sentence comes out. Given the same person and the same day, the app says the same thing forever.

1 · DETECT

Run every condition predicate over chart + pañcāṅga + gocara. Each returns a hit with a weight.

2 · RANK

Sort by weight. Keep the top hit as the subject, the next as the qualifier. Discard the rest.

3 · SELECT

Pick one fragment per slot from the subject's variant list, indexed by a seeded hash.

4 · COMPOSE

Assemble stem + emphasis + close. Mark the emphasis span for the accent tint.

CONDITION WEIGHTS	FRAGMENT SHAPE
Mahādaśā or antardaśā changes100	<pre>{ id: 'budha.direct', weight: 80, when: (c) => c.station('budha','direct'), stem: ['A delayed answer arrives', 'What stalled begins to move'], emphasis: ['the ripe over the new', 'what is already begun'], close: ['Tend it.', 'Say yes slowly.'], suppressIf: ['timeUnknown'] }</pre>
Sāḍhe sātī phase change90	
Retrograde station, benefic or malefic80	
Slow graha ingress to a new rāśi70	
Transit over natal lagna or Moon60	
Candra gocara house from natal Moon40	
Tithi and yoga quality25	
Weekday lord vs. daśā lord10	
Ties break by the graha's natural order. A day with no hit above 25 falls back to the pañcāṅga fragment set, which always yields something.	
SEED <pre>seed = hash(personId + isoDate + fragmentId) index = seed % variants.length</pre> Three variants per slot gives 27 phrasings per condition — enough that the voice does not feel canned, few enough to write well.	

VOICE

Write fragments as observations, not instructions or flattery. Name the sky, then name one thing to do about it. No second-person predictions about money, health or death. Twenty-five to forty words for the long reading; under fifteen for the statement.

Nine screens, five tabs, one push

Flat navigation. Every tab is a root; Paṭha is the only pushed route and it returns to wherever it was opened from. Onboarding is a gate before the shell mounts, not a tab.

ROUTE	SCREEN · DESIGN ID	LOCAL STATE	DATA
/onboard/1	Birth details · 2a	draft BirthData, field validity, geocode result	none
/onboard/2	Confirm cast	computed lagna preview, ayanāṃśa choice	Chart D1
/today	Today · 5a	chip: 'chart' 'transits' 'match' — exclusive, instant swap, panel min-height 196	Reading, Pancanga
/today/read	Full day	scroll only	Reading long
/chart	Natal chart · 2b	varga: D1 D9 chalit; expanded daśā node	Chart, DasaNode
/transits	Gocara · 2c	selected day in week strip; fortnight window offset	GocaraEvent, SadheSati
/match	Guna Milan · 2d	partner id or inline draft; expanded kūṭa row	GunaMilan
/you	Profile · 2e	open settings row; palette picker	Settings, people
/patha	Learn · 3a	entry anchor; pushed, back returns to origin tab	Chart terms

TAB BAR

Five items, top hairline, active in accent, inactive at 50%. 44px minimum hit target. Today · Chart · Transits · Match · You.

TRANSITIONS

Tabs cross-fade at 120ms. Accordion and panel height at 200–250ms ease-out. Chip and segment selection is instant, no animation.

EMPTY AND LOADING

Never a spinner. Hold the layout at its final height with hairlines drawn and text at 0 opacity; charts fade in when the worker replies.

The drawing rules

SIX PALETTES · ACCENT RAMP

Pastel DEFAULT	
Warm gold	
Sindoor	
Twilight	
Rose	
Verdigris	

Each palette supplies bg, surface, text, divider, accent-300/400/600/700/800, ink and ghost. Swatches above show 300 · base · 700 · ink. Store all six in `theme/palettes.ts` and switch by CSS custom properties on the root — never by conditional colour in components.

SPACING

4px base. Screen gutter 24px, hero gutter 26px. Header 14 / 24 / 12. Chip gap 8px. Panel min-height 196px. Kundli 140 on Today, 224 on Chart.

TYPE SCALE

Statement	Cormorant 400 · 27/1.22
Screen title	Cormorant 400 · 30–34
App name	Cormorant 600 · 19 · .04em
Row title	Cormorant 600 · 15–17
CHIP · SEGMENT	Cormorant 600 · 13 · .05em
Body paragraph	Lora 400 · 12.5/1.6
11°04' · 28½/36	Lora 400 · 11 · tabular
KICKER · LABEL	Lora 400 · 9–10 · .14em

NON-NEGOTIABLE

- Square corners everywhere. The device bezel is the only radius in the app.
- Colour arrives as 1px stroke, hairline or text tint. Filled blocks only for an active chip or segment.
- No shadows, no gradients, no elevation. Separation is a hairline at 12% opacity.
- Ghost glyphs sit behind content at 11–16% accent, `pointer-events:none`, never announced to a screen reader.
- All numerals tabular. Degrees as 11°04', scores as vulgar fractions.

Seven milestones, hardest first

The riskiest work is the ephemeris boot and the daśā arithmetic. Do both before any screen exists, verified against the sample persona, so no interface gets built on numbers that turn out wrong.

01	Engine in a worker	no UI
	WASM boots, ephemeris files precached, one promise call returns sidereal graha longitudes. Prove Aisha's Siṃha lagna at 11°04' in a test before writing anything else.	
02	Domain layer with fixtures	no UI
	chart, houses, pañcāṅga, vimśottari daśā, gocara, guṇa milan — each a pure function with a snapshot test. The whole app's correctness is settled here.	
03	Theme, tokens, Kundli	2b, 5a
	Six palettes as custom properties, type scale, and the North-Indian kundli component driven by a Chart. The kundli is the app's signature; get it right once and reuse it at every size.	
04	Onboarding to first chart	2a
	Underline fields, geocoding, the unknown-time fallback, persistence to Dexie, and the shell mounting on a real record. First end-to-end path.	
05	Chart, Transits, Match	2b, 2c, 2d
	The three data screens against real computation. Segmented varga control, week strip and fortnight ledger, aṣṭakūṭa table with the weak kūṭa called out.	
06	Reading engine and Today	5a
	Detectors, weights, the fragment library, then the Today screen with its chip row and swap panel. Today comes last because it summarises everything above it.	
07	Profile, Paṭha, offline	2e, 3a
	Settings ledger, palette picker, saved people, the learn page, service worker and install prompt. Ship.	

DEFERRED TO V2

Local notifications at the chosen reading hour, which needs the Capacitor wrapper. Cloud sync, which the schema already allows. Additional divisional charts beyond D-9.

DEFINITION OF DONE

Airplane mode, cold start, a new birth record cast and read end to end, all six palettes legible, and the fixture tests green.